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Local anesthetic systemic toxicity

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INTRODUCTION

Local anesthetics (LAs) are administered in practice by many clinicians, including anesthesia clinicians, surgeons, emergency department providers, dentists, and others. Despite the widespread use of LAs, awareness of local anesthetic systemic toxicity (LAST) and knowledge of its management are lacking [1,2].

Local anesthetics are often incorrectly thought to be without side effects or toxicity. LAST is always a potential complication, and may occur with all LAs, and any route of administration. LAST primarily affects the central nervous system and cardiovascular system, and may be fatal.

This topic will discuss the pathologic pharmacology, mechanisms, and clinical presentation of toxicity, predisposing factors, and prevention and management of LAST. Lipid rescue therapy will also be discussed.

INCIDENCE OF LAST

The reported incidence of major LAST events (ie, seizures or cardiac arrest) associated with regional anesthesia is very low. Major LAST events have decreased markedly since the early 1980s, likely related to increased awareness and routine incorporation of preventive measures into clinical practice [3,4]. (See 'Prevention of LAST' below.)

Data regarding the incidence of LAST come from registry studies and retrospective reviews of large numbers of regional anesthetics, based on very small numbers of events. Major LAST events are reported in 0.04 to 1.8/1000 peripheral nerve blocks. It should be noted that the epidemiology of LAST is likely changing, given the widespread use of ultrasound and

increased use of novel fascial plane blocks since the studies below were published. Examples of relevant studies include the following:

- In one administrative database study, data was reviewed for 238,437 patients who received a peripheral nerve block for total joint arthroplasty between 2006 and 2014 [5]. The overall incidence of LAST (defined as occurrence of cardiac arrest, seizure, and/or administration of lipid emulsion on the day of surgery) was 1.8 per 1000. The incidence of the composite outcomes declined over time (from 8.2 per 1000 in 2006 to 2.5 per 1000 in 2014), whereas the administration of lipid emulsion increased from 0.2 to 2.6 per 1000 over the same time period.

- In another review, data from the National Inpatient Sample in the United States was used to identify 710,327 patients who underwent total joint arthroplasty with an adjunctive peripheral nerve block between 1998 and 2013 [6]. The incidence of LAST (identified by billing codes) was 1.04 per 1000, with major complications (ie, seizure or cardiac complications) in 21 percent of patients with LAST. The odds of LAST decreased by 10 percent per year over the study period.

- A retrospective single institution analysis of approximately 80,600 peripheral nerve blocks from 2009 to 2014 reported three cases of seizures caused by LAST (0.04 per 1000) and no cases of cardiac arrest [7].

- A review of data from the multicenter Australian and New Zealand Registry of Regional Anesthesia (AURORA) included data from approximately 25,300 peripheral nerve blocks performed from 2008 to 2012 [8]. The overall incidence of minor and major LAST events was 0.87 per 1000 blocks. Major LAST events occurred in eight patients (0.31 per 1000), including seven cases of major central nervous system events (ie, seizure or unconsciousness) and one case of cardiac arrest. Ultrasound guidance (USG) was associated with reduced risk of LAST (0R 0.36 for minor and major LAST events).

Other reviews have reported similarly low incidences of major LAST events [9-13]. The incidence of major LAST may be higher with peripheral nerve block than with epidural block [9], but this conclusion is based on very small numbers of events. Minor LAST events (eg, tinnitus, perioral numbness, metallic taste) probably occur much more frequently than major events, and go unreported.

PHARMACOLOGY AND PATHOPHARMACOLOGY OF LAST

All local anesthetics have a hydrophobic aromatic ring and a hydrophilic amine group. The intermediary chain may contain either an ester or amide link, and dictates metabolism via plasma cholinesterases or hepatic oxidation, respectively. Most local anesthetics, particularly amide-type, are chiral molecules and therefore have two enantiomers [14]. These

enantiomers may be designated in terms of its spatial orientation (R- for clockwise and S- for counterclockwise configuration) or by which direction it rotates plane-polarized light (D- or dextro for the right and L- or levo for the left) [15]. The chirality of these compounds account for some of their subtle differences in clinical effects [16]. For example, bupivacaine exists as a racemic mixture, while levobupivacaine and ropivacaine are pure (S-) enantiomers [16,17]. Though clinically similar, slight differences in anesthetic duration (bupivacaine > levobupivacaine > ropivacaine) is attributed to stereochemistry.

The physiochemical properties of local anesthetics are responsible for individual pharmacokinetic differences among the clinically used agents [18].

- The **pKa** and its relationship to the surrounding pH influence the onset of effect due to the proportion of charged ionized drug to its uncharged un-ionized form [18]. Lower pKa indicates that a greater proportion exists in the uncharged state, so more drug is available to transfer across the lipophilic cellular membrane to its binding site on the intracellular surface of the sodium channel. (See "Subcutaneous infiltration of local anesthetics", section on 'Anatomy and physiology'.)

- Similarly, **lipophilicity** correlates to potency, because a higher lipophilicity confers better ability to penetrate the lipid bilayer to the target [18].

- **Protein binding** of the drug primarily to alpha-1-acidic glycoprotein, and to a lesser extent albumin, and tissue proteins relates to effect duration, serving as a reservoir for local anesthetic. A high affinity for protein binding decreases the effective circulating levels of free local anesthetic [18].

All local anesthetics exert their effect primarily by blocking voltage-gated sodium channels (VGSC) at the alpha-subunit inside the channel, which prevents sodium influx, subsequent depolarization, and action potential generation [19]. This conduction block impedes pain transmission from neuronal cells to the cerebral cortex, thereby producing analgesia and anesthesia. Toxicity occurs when LAs affect cardiac sodium channels [20,21] or thalamocortical neurons in the brain [22]. LAs have other, wide ranging effects, including blockade of potassium and calcium channels, interaction with cholinergic or N-methyl-D-aspartate (NMDA) receptors, and interference with cellular metabolic processes (eg, oxidative phosphorylation, free fatty acid utilization, cyclic adenosine monophosphate production). Such effects may explain some aspects of cardiovascular toxicity, and may relate to the mechanism for reversal of LAST with intravenous lipid emulsion (ILE). (See 'Mechanism of action' below and 'Clinical presentation of toxicity' below.)

Differences in toxicity among local anesthetics — All local anesthetics (LAs) may cause LAST with toxic plasma levels. However, cardiac toxicity and the relationship between central nervous system (CNS) and cardiac toxicity vary among LAs. Cardiac toxicity parallels the

anesthetic potency for neural blockade [23]; the more potent local anesthetics, including etidocaine and bupivacaine, are more cardiotoxic than lidocaine.

Amide local anesthetics — Ropivacaine and levobupivacaine, which are prepared in almost pure S isomer form, are slightly less cardiotoxic [24,25] and produce fewer CNS symptoms [26] than bupivacaine. The use of ropivacaine and levobupivacaine may help reduce cardiotoxicity, but does not eliminate the potential for LAST, especially at higher doses.

LAs can be assigned a cardiovascular collapse (CC) to CNS ratio (CC:CNS), which is the ratio of the drug dose required to cause cardiac collapse to the drug dose required to produce seizures. LAs with higher CC:CNS theoretically have a greater margin of safety, as early CNS symptoms may allow intervention to prevent subsequent cardiac arrest. Based on animal models, bupivacaine has the lowest CC:CNS ratio of the commonly used LAs, followed in increasing order by levobupivacaine, ropivacaine, and lidocaine [27].

LAs may have specific cardiac effects; lidocaine and mepivacaine primarily affect contractility, while bupivacaine and ropivacaine tend to be more arrhythmogenic [28].

Ester local anesthetics — Chloroprocaine is a short acting ester LA that is metabolized by plasma cholinesterases, and may be used for epidural anesthesia and peripheral nerve blocks. It's very short plasma half-life may confer a degree of safety if inadvertent intravascular injection or systemic absorption occurs. The few LAST events that have been reported with chloroprocaine have been very brief. In single case reports, a 30 second seizure after paravertebral block [29] and 40 second cardiac dysrhythmia after inadvertent intravascular injection during epidural block [30] have been reported with chloroprocaine in infants. Minor LAST events lasting 3 to 11 minutes have also been reported after intravenous (IV) regional anesthesia using chloroprocaine [31].

Plasma half-life of chloroprocaine may be prolonged in patients with pseudocholinesterase deficiency [32] and in patients who take anticholinesterase medication (eg, pyridostigmine for myasthenia gravis) [33], and presumably such patients are at risk for more prolonged LAST events. Clinicians should consider using a lower dose or an alternative local anesthetic, and should maintain a high level of clinical suspicion for toxicity when administering chloroprocaine in such patients. (See "Clinical use of local anesthetics in anesthesia", section on 'Metabolism of local anesthetics' and "Anesthesia for the patient with myasthenia gravis or Lambert-Eton myasthenic syndrome", section on 'Choice of anesthetic technique'.)

ROUTES OF LOCAL ANESTHETIC ADMINISTRATION

LAST may occur after any route of administration of local anesthetic (LA).

- **Regional anesthesia** – The majority of reported cases of LAST have occurred after inadvertent intravenous (IV) injection of LA. However, systemic absorption from LA administration sites can also cause toxic blood levels, and may cause a delayed, gradual onset of symptoms and signs. Continuous LA infusion for regional analgesia can cause LAST by intravascular migration of the catheter or systemic absorption, and onset may occur hours or days after initial catheter placement. Rarely, LAST can occur after inadvertent low dose intraarterial injection during nerve block in the neck (eg, during interscalene block, cervical plexus block, or stellate ganglion block). In this setting, seizure usually occurs without cardiovascular effects, and usually resolves quickly [34-37].

- **Topical application** – LAST can also occur after topical administration of LAs, especially when the drug is applied to mucosal membranes in repeated or high doses. Lidocaine used for topical oropharyngeal anesthesia is absorbed through mucous membranes, and from the gastrointestinal tract when swallowed. There are multiple case reports of seizures and cardiac arrest with high doses of oral viscous lidocaine, in adults [38,39] and children [40] (see 'Patient risk factors' below). LAST has been reported in children after inappropriate ingestion of other local anesthetic-containing preparations, such as topical analgesics that contain dibucaine [41].

  High blood levels of lidocaine have been measured after topical administration for bronchoscopy [42,43], and cases of cardiac arrest have been reported [44-46].

  Topical application of LAs to the skin can also result in high blood levels and toxicity. LAST has been reported after the use of EMLA cream [47] and topical tetracaine in children [48]. Lidocaine applied to the face may result in high and variable blood levels, especially when used with an occlusive dressing [49].

RISK FACTORS FOR LAST

Risk factors for LAST include patient factors that increase the free plasma drug concentration, or increase sensitivity to local anesthetics (LAs). Systemic absorption is increased at highly vascular block sites, independent of the drug or dose injected. In addition to total dose per body weight, these patient and procedural factors may significantly affect plasma levels in adults and therefore, contribute to increased risk for toxicity [50,51].

Patient risk factors — Free LA in plasma may be increased by a low alpha 1 acid glycoprotein (AAG) level [52]. AAG binds to LA in the plasma so there is an inverse relationship between their respective levels in the plasma. In addition, patients with hyperdynamic circulation (eg, pregnancy, uremia) can rapidly absorb LA from injection sites, and plasma levels can rise quickly. Plasma LA levels after repeat dosing or infusion can be increased by impaired hepatic clearance.

- **Extremes of age** – Infants under the age of approximately four months have low levels of AAG, and may be at increased risk of LAST [52,53]. They may also have immature hepatic clearance, and doses for repeat or continuous administration should be reduced by 15 percent from published standard doses for other ages [54,55].

  Infants and young children may be at increased risk for LAST because of their small size, particularly with topical administration. As an example, it is easy to administer too large a dose of LA in what seems to be a small amount of LA gel. The US Food and Drug Administration (FDA) has issued a warning about the risks of severe LAST or death with the use of viscous lidocaine 2% in young children.

  In older adult patients, clearance of local anesthetics may be reduced due to deterioration of hepatic function and perfusion [56]. Thus they may be at increased risk for LAST with repeat dosing or continuous infusion of LAs [57]. In addition, nerves are more sensitive to LAs in older adults, and lower doses of LA are necessary for adequate block than in younger patients [58].

- **End organ dysfunction** – Generally, end organ dysfunction does not necessarily require dose adjustment, unless it is in its advanced stages.

  - **Cardiac disease** – Many cases of LAST have occurred in patients with underlying ischemic or other cardiac disease [34], but it is not clear that cardiac disease actually predisposes to LAST. Heart failure severe enough to affect hepatic and renal perfusion can reduce LA clearance, and increase the risk of LAST with repeat or continuous dosing [59].

    Patients with ischemic heart disease or heart failure may be more difficult to resuscitate if cardiac toxicity does occur.

  - **Renal insufficiency** – Renal dysfunction does not increase risk for toxicity, unless metabolic acidosis ensues. The hyperdynamic circulation of uremic patients causes a rapid rise in LA plasma levels after large volume nerve block, but their high levels of AAG reduce the level of free drug [60].

  - **Hepatic disease** – Hepatic dysfunction itself does not increase the risk of toxicity with single injection nerve blocks, unless liver disease is end stage, or associated with other comorbidities. Reduced LA clearance is offset by a larger volume of distribution for the initial injection and alpha-1-acid glycoprotein synthesis is usually preserved despite severe disease [49,61]. However, severe liver disease increases the risk of LAST with continuous infusion or repeat block, and doses of LA should be reduced.

- **Pregnancy** – Pregnant patients, especially at term, are at increased risk for LAST, for several reasons:

- Hormonal changes may increase sensitivity of neural tissue to block [62], and to cardiotoxicity [63].
- Reduced levels of AAG and albumin may increase the free fraction of some LAs in pregnant patients [64].
- The increased cardiac output of pregnancy may cause rapid absorption of LA after injection. Epidural venous engorgement may also increase absorption of LA during epidural anesthesia.
- Lower pseudocholinesterase activity during pregnancy may reduce chloroprocaine metabolism and theoretically increase the duration of a LAST event if it occurs. (See 'Ester local anesthetics' above.)

The FDA issued a warning for the use of bupivacaine, which states that bupivacaine 0.75% is not recommended for obstetrical anesthesia. The warning is based on reported cases of cardiac arrest with difficult resuscitation or death in obstetric patients who received bupivacaine for epidural anesthesia. Most of these cases involved administration of 0.75% bupivacaine, but LAST may occur with any concentration of LA.

- **Others** – Metabolic disturbances such as acidosis, hypoxia, hypercarbia may increase the risk of toxicity [65]. Carnitine deficiency also increases risk for cardiotoxicity, particularly with bupivacaine [66-68].

Block site — Injection of LA in highly vascular sites can increase the risk of direct intravascular injection and systemic absorption of LA. Regional anesthesia procedures that target particularly vascular sites include (in order of decreasing risk) intercostal blocks> caudal and epidural anesthesia> interfascial plane blocks of the abdominal wall> psoas compartment blocks> sciatic blocks> and cervical and brachial plexus blocks [69,70].

Blocks that require large volumes and doses of local anesthetic may increase the risk of systemic absorption [71]. As an example, interfascial plane blocks (eg, transversus abdominis plane [TAP] block) are often performed bilaterally, and involve injection of a total of ≥20 mL of LA per side to ensure adequate spread. Potentially neurotoxic plasma concentrations of LA have been reported in patients who received TAP blocks prior to laparotomy [72,73], and neurotoxicity, including seizures, has been reported with TAP blocks in patients who are at increased risk of LAST (eg, renal or hepatic disease) [74,75], and in parturients who received TAP block for pain relief after cesarean delivery [76,77]. (See "Transversus abdominis plane (TAP) blocks procedure guide", section on 'Complications'.)

PREPARATION AND MONITORING FOR LAST

Emergency resuscitation equipment should be available whenever local anesthetic is injected, to allow management of LAST and other complications of local anesthetic (LA) administration (eg, vagal response). Prior to performance of regional anesthesia (ie, peripheral nerve block, neuraxial block) or airway topicalization, intravenous (IV) access should be established, and monitors should be applied, including continuous pulse oximetry and electrocardiogram, and blood pressure monitoring.

We follow the American Society of Regional Anesthesia and Pain Medicine (ASRA) safety checklist prior to performance of regional nerve blocks (table 1) [78]. The ASRA checklist incorporates features of the World Health Organization preanesthesia safety checklist (table 2), and includes a check for availability of resuscitation equipment, including vasoactive drugs and lipid emulsion.

We administer supplemental oxygen to patients during performance of peripheral nerve blocks. We continue physiologic monitoring for at least 30 minutes after local anesthetic administration, to detect signs of delayed presentation of LAST.

PREVENTION OF LAST

A number of techniques are used to reduce the risk of LAST, including utilizing the lowest effective dose, safe injection techniques, the use of ultrasound, and avoidance of heavy sedation.

Local anesthetic dose — The total dose of local anesthetic administered should be the lowest dose that is required for the desired extent and duration of block. Maximum allowable doses that appear in various publications are rough guidelines that are not evidence based, and don't take into account the site or technique of local anesthetic (LA) administration, or patient factors that increase the risk of toxicity. Systemic toxicity may occur with doses within the recommended ranges, while doses in excess of the recommended maximums have been administered without toxicity [26,79]. Nonetheless, published maximum recommended doses can be used as a starting point for deciding on a dose (table 3). The following general considerations should be followed (see 'Patient risk factors' above):

- Maximum dose of local anesthetic should be based on lean, rather than actual body weight (calculator 1 and calculator 2).
- Dose of LA should be reduced for patients who are at risk for rapid uptake of LA from injection site (eg, pregnancy, uremia).
- Dose of LA should be reduced for patients with low levels of alpha 1 acid glycoprotein (eg, neonates, pregnant patients), who are at risk for high concentrations of free LA after injection.

- Dose of LA should be reduced for patients who have increased neural sensitivity to LA (eg, older adults, pregnant patients).

- Dose of local anesthetic for continuous or repeat administration should be reduced for patients at risk for reduced clearance of LA (eg, renal, hepatic, or cardiac disease).

- Addition of epinephrine to LA solutions can slow the rate of absorption and reduce peak plasma levels [80], and may act as a marker for intravascular injection. Epinephrine reduces the absorption of LA by 20 to 50 percent, depending on the site of injection and the LA [50,80-82].

Safe injection techniques — We use the following injection techniques for regional anesthesia, to reduce the risk of intravascular injection and for early detection of accidental intravascular injection:

- **Slow incremental injection** – We administer LA incrementally, (eg, 5 mL aliquots), with at least 30 to 45 seconds between injections, and with slower injection with smaller aliquots for more concentrated LA solutions. This delayed injection technique should allow for detection of accidental intravascular injection before toxic doses are given [20].

- **Aspiration prior to injection** – We aspirate gently prior to each incremental injection, and reposition the needle or catheter if we see blood return. Falsely negative aspiration is possible. As an example, in parturients, false negative aspiration with multi-orifice epidural catheters may be as high as 2 percent [83-85].

- **Intravascular marker** – We add epinephrine (1:200,000 or 1:400,000) to most LA solutions (or used premixed epinephrine containing solutions) to act as a test dose for intravascular injection, in addition to the other effects of epinephrine. Intravascular injection of 15 mcg epinephrine usually results in an increase in heart rate of ≥10 beats per minute, and increase systolic blood pressure ≥15 mmHg within 20 to 40 seconds [86], and is considered a positive test dose. Beta blockade, labor pain, advanced age, heavy sedation, and general anesthesia may confound the effects of epinephrine. (See "Neuraxial analgesia for labor and delivery (including instrumental delivery)", section on 'The epidural test dose in obstetrics'.)

  For patients who are under general anesthesia, an increase in systolic blood pressure after an epinephrine containing test is a more reliable indicator of intravascular injection than an increase in heart rate, both during inhalation anesthesia, and total intravenous (IV) anesthesia [87-89].

Avoidance of heavy sedation or general anesthesia — We avoid heavy sedation during performance of regional anesthesia techniques, and perform such techniques under general anesthesia only when absolutely necessary. Patient feedback may help detect nerve injury

(eg, by complaint of pain or paresthesia), progression of motor and sensory block, and early signs of LAST. Without appreciation of minor LAST symptoms (eg, tinnitus, circumoral numbness, dizziness), cardiovascular toxicity may occur as the first sign of LAST [34].

Regional anesthesia is routinely performed under general anesthesia for pediatric patients, and the risk of LAST appears to be very low when safe injection techniques are used. In a review of approximately 15,000 regional blocks reported to the Pediatric Regional Anesthesia Network, which is a multicenter collaborative network, there were no cases of LAST, though there were 13 cases of positive test dose, and 33 cases of vascular puncture [90]. Ninety-five percent of blocks were placed during general anesthesia. Multicenter reviews of pediatric regional anesthesia in Europe have also reported very low rates of LAST [91,92].

Ultrasound guidance — We use ultrasound guidance (USG) for most peripheral nerve blocks, to improve the safety and efficacy of these techniques.

USG may decrease the risk of LAST by several mechanisms:

- Reduced vascular puncture by visualization and avoidance of vascular structures, or by manual compression of vessels by the ultrasound transducer [8,93]
- Use of visualized spread of LA solution as confirmation of extravascular injection [94]
- Reduced total LA dose, compared with landmark based or nerve stimulation-based techniques [95,96]

The impact of USG on reducing LAST is debated. Large databases include very small numbers of LAST events, and randomized trials are too small to show an effect of block technique on LAST. As examples, a meta-analysis of 13 randomized controlled trials including approximately 950 patients that compared USG with peripheral nerve stimulation found that USG decreased the risk of vascular puncture during performance of peripheral nerve block (RR0.16), but studies were too small to draw conclusions about effects on LAST [93]. A review of over 25,000 blocks in a multicenter registry database (AURORA) [8], and a review of over 9000 blocks in a single institution database [13] both found that USG reduced the incidence of LAST. However, the total number of major LAST events was small (14 in approximately 34,000 blocks between the two studies). Thus, definitive conclusions regarding a reduction in LAST with USG cannot be drawn. Regardless, major LAST events have been reported during USG, and vigilance should be maintained during any LA injection.

CLINICAL PRESENTATION OF TOXICITY

The clinical presentation of LAST is highly variable, and LAST should be suspected whenever physiologic changes occur after administration of local anesthetic (LA). Descriptions of the

clinical presentation come from older studies, and may differ in current practice with the widespread use of ultrasound and use of novel fascial plane blocks.

- **Onset** – As classically described in the literature and based on older studies, LAST occurs soon after injection of LA, and progresses through central nervous system excitation, central nervous system (CNS) inhibition, cardiovascular excitation, and in extreme cases, cardiovascular inhibition and arrest. However, in a review of 93 reported LAST events that occurred during regional anesthesia over 30 years until 2009, only 60 percent of reported cases followed this classic presentation [34] (figure 1). In the remaining cases, onset was delayed or involved only cardiovascular (CV) effects, without CNS toxicity. In a subsequent review of 67 cases of LAST reported between 2010 and 2014, the distribution of signs and symptoms was similar to the previously reported cases, but a much greater proportion had an onset >10 minutes after injection (42 versus 13 percent), and 22 percent occurred >30 minutes after injection (figure 2 and figure 3 and figure 4 and figure 5) [37]. Similarly, in a review of 47 reported cases of LAST between 2014 and 2016, onset occurred after 10 minutes of LA injection in approximately 43 percent of cases, and approximately 23 percent occurred after 60 minutes [97]. LAST occurred most commonly as a result of penile blocks in infants (23 percent of reports).

  The presentation and speed of progression of symptoms depend on the route of systemic absorption, the local anesthetic plasma level, how quickly the concentration rises, and the characteristics of the specific LA. (See 'Differences in toxicity among local anesthetics' above.)

- **Central nervous system effects** – Local anesthetic blood levels in the brain initially block cortical inhibitory pathways, and may therefore cause excitatory signs and symptoms, including perioral numbness, metallic taste, mental status changes or anxiety, visual changes, muscle twitching, and ultimately, seizures. The more minor of these effects may occur with intravascular injection of a small dose of LA (eg, after an epidural test dose, or an incremental dose of LA during peripheral nerve block), or early after injection of larger doses. These symptoms and signs should alert the clinician to stop injection, and prepare for progression to more serious CNS and/or cardiovascular events (figure 3). (See 'Safe injection techniques' above.)

  Increasing blood levels then cause generalized CNS depression, which can result in somnolence, coma, and respiratory depression.

- **Cardiovascular effects** – CV signs and symptoms usually occur after or along with CNS symptoms during LAST, though they may occur alone. Initial sympathetic activation can cause tachycardia and hypertension. However, bradycardia and hypotension have often been described as the first changes in vital signs [34]. CV toxicity can progress to ventricular arrhythmias and/or asystole (figure 2).

The CV effects of systemic LA are complex and multifactorial, and are not completely understood. In laboratory studies, LA can block sodium, calcium [98], and potassium channels [99-102], and can cause conduction disturbances, impaired cardiac contractility, and loss of vascular tone.

MANAGEMENT OF LAST

Our approach to the management of LAST is consistent with the guidelines and checklists published by the American Society of Regional Anesthesia and Pain Management (ASRA) [103,104] and the Association of Anaesthetists of Great Britain and Ireland (AAGBI) [105] (table 4). Key components of successful management include rapid response, maintenance of oxygenation and ventilation, seizure suppression, and cardiovascular support. Once LAST is suspected, the following steps should be followed:

- **Stop injection** when any signs of intravascular injection occur.
- **Call for help** – Call for qualified assistance, a LAST toxicity kit or lipid emulsion, and any available cognitive aid (eg, emergency manual), if progressive or severe signs or symptoms of LAST occur. Confirm or establish intravenous (IV) access.
- **Arrange for cardiopulmonary bypass** – Alert the nearest facility with cardiopulmonary bypass (CPB) capability. Arranging for CPB takes time, and though it may not be necessary, CPB may be the last resort, and life-saving. CPB may be necessary while waiting for local anesthetic (LA) to diffuse from cardiac receptors, if advanced cardiac life support (ACLS) and IV lipid emulsion (ILE) fail.
- **Manage the airway** – Administer 100 percent oxygen by face mask, and if necessary, control ventilation with a face mask, supraglottic airway, or endotracheal tube. The goal should be to prevent hypoxia and acidosis, as both potentiate LAST.
- **Suppress seizure** – Suppress seizures immediately to reduce oxygen consumption, prevent hypoxia and hypercarbia, and prevent patient injury. Administration of a benzodiazepine is preferred (eg, midazolam 1 to 2 mg IV). Alternatively, small doses of propofol (eg, 20 mg increments) or thiopental (where available) may be used cautiously, as these drugs can cause hypotension and may exacerbate cardiovascular effects of LAST. If necessary, administer succinylcholine to terminate tonic-clonic muscle activity and attenuate metabolic acidosis, recognizing that this will have no effect on central nervous system electrical activity.
- **Manage arrhythmias and provide cardiovascular support** – Management of arrhythmias and cardiac arrest as a result of LAST is different than other cardiac arrest scenarios, and may require prolonged effort. A goal is to maintain coronary perfusion and prevent cardiac

tissue hypoxia. High quality chest compressions must be used during cardiac arrest, to perfuse the coronary arteries and to circulate lipid emulsion, which is usually administered simultaneously.

Advanced cardiac life support principles should be followed, with the following modifications:

- Administer amiodarone as the first line antiarrhythmic (usual doses per ACLS), and avoid lidocaine and other sodium channel blockers (eg, other LAs, Class 1 antiarrhythmics [quinidine, procainamide, disopyramide, mexiletine, tocainide], antiepileptic drugs).
- Reduce the bolus dose of epinephrine to ≤1 mcg/kg IV to avoid arrhythmogenic effects. This recommendation is based on animal studies [106,107], and the optimal dose of epinephrine is unknown.
- Avoid vasopressin. In an animal model of bupivacaine induced cardiac arrest, vasopressin was associated with poor hemodynamic and metabolic outcomes [108].
- Avoid calcium channel blockers and beta blockers because of potential for exacerbating hypotension.

- **Administer lipid rescue** – Precise volumes and rate of administration of lipid emulsion are not crucial. Administer 20 percent lipid emulsion along with advanced cardiac life support or when neurotoxicity occurs, as follows (see 'Lipid rescue' below):
  - Bolus lipid emulsion rapidly over two to three minutes and begin infusion based on patient ideal body weight as follows:
    - **<70 kg** – 1.5 mL/kg IV, followed by infusion at 0.25 mL/kg/minute IV
    - **≥70 kg** – 100 mL IV, followed by infusion of 250 mL IV over 15 to 20 minutes.
  - For patients with persistent cardiovascular instability, repeat the bolus and double the infusion rate. Continue infusion for at least 15 minutes after hemodynamic stability is achieved.
  - Maximum dose of lipid emulsion is approximately 12 mL/kg
  - Propofol is not a substitute for 20 percent lipid emulsion as it provides very little lipid per mg of propofol.
  - Monitor the patient for two hours after seizure, four to six hours after hemodynamic instability, or as appropriate after cardiac arrest.
- **Transfer patient to monitored setting** – Upon successful stabilization, the patient should be safely and expeditiously transferred to a monitored setting that is staffed and equipped

to continue recovery [109]. Though it would be very unusual to occur, vigilance should still be maintained for possible recrudescence of LAST [110]. The patient should be followed for any complications or side effects. And finally, cases of LAST or suspected LAST should be reported to the appropriate national registries for tracking [109].

LIPID RESCUE

We recommend administration of intravenous lipid emulsion (ILE) for patients with LAST who exhibit seizures or impending seizures, or signs of cardiovascular toxicity (ie, arrhythmias, severe hypotension, or cardiac arrest) along with ACLS. Though the optimal lipid preparation, dose of administration, and mechanism of action are unclear, LAST may be fatal, and adverse effects appear to be rare.

Efficacy — Numerous case reports describe successful resuscitation of cardiac arrest related to LAST by intravenous (IV) infusion of lipid emulsion after patients failed to respond to standard advanced cardiac life support (ACLS) [111-113]. Local anesthetics reported in these cases include lidocaine, mepivacaine, ropivacaine, levobupivacaine, and bupivacaine, used as sole agents or in combination. A systematic review of both human and animal studies involving IV lipid emulsion (ILE) therapy for LAST concluded that the published evidence was limited, and that ILE may be effective for reversal of cardiac or neurologic sequelae in some cases of LAST [111]. There was no consistent evidence to support ILE therapy over vasopressors, or to determine whether one treatment should precede the other. However, this systematic review itself was severely limited by the lack of randomized controlled human trials, a heavy reliance on animal studies, and heterogeneity in study design, animal model utilized, and reported results.

Mechanism of action — The mechanism of action of lipid emulsion for treatment of LAST is unclear, and may be multifactorial, based on laboratory studies. Historically, lipid was thought to act as a "lipid sink," whereby lipid would bind the local anesthetic (LA) to remove it from target tissue. A somewhat different emerging theory is that lipid emulsion carries or "shuttles" the LA from the heart and brain to the organs that store and detoxify the drug [114-118]. Lipid emulsion may also have direct cardiotonic effects involving sodium channels [119], fatty acid processing [120,121], and mitochondrial metabolism or permeability [122], which would enhance the shuttling effect.

Timing of administration — The timing of lipid rescue therapy is controversial. We recommend administration of lipid emulsion immediately for patients with LAST who exhibit seizures or impending seizures, or signs of cardiovascular toxicity (ie, arrhythmias, severe hypotension, or cardiac arrest). We agree with the guidelines from the American Society for Regional Anesthesia and Pain Medicine (ASRA) [103], the American Heart Association (AHA) [123], and the Association of Anaesthetists of Great Britain and Ireland (AAGBI) [105] that lipid

emulsion should be administered concurrently as part of ACLS, if cardiac arrest is suspected to be a result of LAST.

Available formulations — The optimal lipid preparation for treatment of LAST is debated [124]. The formulation most commonly used has been 20 percent lipid emulsion (eg, Intralipid, Liposyn III 20%, or Nutrilipid) which is widely available in hospital formularies for nutritional support. Some formulations are also available in 10 and 30 percent concentrations, and animal studies suggest that at least some aspects of lipid rescue may be dose dependent [125]. Further study is required before recommending concentrations of lipid emulsion other than 20 percent for treatment of LAST.

Propofol is prepared in 10 percent lipid emulsion. Propofol is never a suitable alternative to 20 percent lipid emulsion because of its cardiovascular depressant effects, particularly with the high volumes that would be required to treat LAST.

The commercially available forms of 20 percent lipid emulsion contain soybean oil and egg yolk phospholipids, and should not be used in patients with true allergies to either of these food substances. The manufacturers state that lipid emulsion should be administered by IV infusion only, but intraosseous administration has been reported in a case in which IV access was lost [126].

Dose — The optimal dose of lipid emulsion for LAST has not been determined. Multiple organizations have published guidance for lipid rescue therapy, including the ASRA [103],the AAGBI [105], and the American College of Medical Toxicology (ACMT) [127]. Our approach is consistent with 2020 guidance from ASRA. All recommendations use 20 percent lipid emulsion. ASRA recommends a dose of 100 mL over two to three minutes for patients >70 kg followed by an infusion of 250 mL over 15 to 20 minutes, and a bolus of 1.5 mL/kg for patients <70 kg followed by a continuous infusion of 0.25 mL/kg/minute. The other organizations recommend an initial bolus of 1.5 mL/kg for all patients followed by infusion of 0.25 mL/kg/minute. ASRA recommends repeat bolus as necessary for persistent or recurrent cardiovascular instability.

These organizations have issued somewhat different recommendations regarding administration of lipid emulsion after the initial bolus and infusion as follows:

- ASRA [103,104] – Continue infusion for at least 15 minutes after hemodynamic stability has been restored. Repeat the bolus and increase the infusion to 0.5 mL/kg/minute if hemodynamic instability persists or recurs. Maximum total dose should be 12 mL/kg. Note that there is little data to support a maximum of 12 mL/kg, however it is unlikely that more than 12 mL/kg will be beneficial.

- AAGBI [105] – Continue infusion until stable or maximum dose given, 12 mL/kg. Give up to two repeat bolus doses, five minutes apart, and double infusion to 0.5 mL/kg/minute at any

time after five minutes, for persistent or recurrent cardiovascular instability.

- ACMT [127] – If there is an initial response, followed by re-emergence of hemodynamic instability, the infusion rate may be increased or the bolus repeated. Where possible lipid therapy should be discontinued after one hour or less. Longer treatment may be required.

There is no published guidance on dosing of lipid emulsion for LAST associated with liposomal bupivacaine, a sustained release formulation of bupivacaine. The initial dose of lipid emulsion should presumably be the same as for other local anesthetics. The concern would be the possibility of prolonged LAST effects, as the peak plasma levels of bupivacaine can occur as long as 36 hours after injection of liposomal bupivacaine, with wide variability depending on injection site. Patients with LAST after liposomal bupivacaine may require prolonged infusion of lipid emulsion, and/or observation in a monitored setting. Liposomal bupivacaine is discussed in detail separately. (See "Clinical use of local anesthetics in anesthesia", section on 'Liposomal bupivacaine'.)

Adverse effects — The safety of high volume lipid infusion for LAST is not precisely known. The most commonly reported adverse effect of lipid rescue has been the interference with laboratory testing, including coagulation testing [128], which may last for several hours after lipid administration [129-131]. Centrifugation of blood samples substantially reduces interference [128,132].

Rare cases of pancreatitis and deep vein thrombosis have also occurred. Lipid emulsion may interfere with filters used for renal replacement therapy [133,134], and importantly, may cause fat deposition and blood clots in cardiopulmonary bypass and extracorporeal membrane oxygenator circuits [135].

Other therapeutic uses for lipid rescue — Applications for lipid rescue include toxicity associated with a number of other lipophilic medications, including antiarrhythmics, anticonvulsants, antidepressants, antipsychotics, benzodiazepines and benzodiazepine-agonist, calcium channel blocker, cocaine, and diuretics [136,137]. (See "Beta blocker poisoning", section on 'Lipid emulsion therapy' and "Calcium channel blocker poisoning", section on 'Lipid emulsion therapy'.)

CRISIS MANAGEMENT

LAST is a rare, potentially fatal event that requires a rapid multifactorial response. Similar to other critical incidents, the use of cognitive aids (eg, checklists) and simulation training are important components of optimal preparation and management. The uses of cognitive aids and simulation for perioperative emergencies are discussed separately. (See "Cognitive aids for perioperative emergencies" and "Patient safety in the operating room", section on 'Team and simulation training'.)

SOCIETY GUIDELINE LINKS

Links to society and government-sponsored guidelines from selected countries and regions around the world are provided separately. (See "Society guideline links: Treatment of acute poisoning caused by specific agents other than drugs of abuse" and "Society guideline links: Local anesthetic systemic toxicity".)

SUMMARY AND RECOMMENDATIONS

- **Rapid overview** – The risk factors, clinical manifestations, and treatment of local anesthetic systemic toxicity (LAST) is shown in a table (table 4).
- **Importance of LAST** – LAST affects the central nervous system (CNS) and the cardiovascular system. While the reported incidence of major LAST events (eg, seizure, cardiac arrest) associated with regional anesthesia is very low (0.04 to 1.8 per 1000), such events may be fatal. (See 'Incidence of LAST' above.)
- **Differences among local anesthetics** – All local anesthetics have the potential to cause LAST. Cardiac toxicity and the relationship between cardiovascular (CV) toxicity and central nervous system (CNS) toxicity vary among local anesthetics. Bupivacaine is the most cardiotoxic of the commonly used LAs, followed in decreasing order by levobupivacaine, ropivacaine, and lidocaine. (See 'Differences in toxicity among local anesthetics' above.)
- **Risk factors**
  - Patient factors that increase the risk of LAST include extremes of age, renal, hepatic or cardiac disease, pregnancy, and metabolic disturbances. (See 'Patient risk factors' above.)
  - Procedural risk factors include highly vascular block sites and blocks that require large volumes of local anesthetics. (See 'Block site' above.)
- **Prevention** – The risk of LAST during regional anesthesia may be reduced by the following (see 'Prevention of LAST' above):
  - Limit the dose of local anesthetic to the lowest effective dose
  - Add epinephrine 1:200,000 to 1:400,000 to the local anesthetic solution as an intravascular marker
  - Aspirate prior to each injection
  - Inject small incremental doses (eg, 3 to 5 mL), assessing for signs of LAST between injections

- Use ultrasound guidance for peripheral nerve blocks
- Avoid heavy sedation or general anesthesia during block procedures

- **Clinical presentation** – The clinical presentation of LAST is highly variable (figure 1). (See 'Clinical presentation of toxicity' above.)
  - **CNS** – LAST events often start with signs and symptoms of CNS excitation (eg, perioral numbness, metallic taste, mental status changes or anxiety, visual changes, muscle twitching, and ultimately, seizures), followed by CNS depression (eg, somnolence, coma, and respiratory depression).
  - **CV** – CV signs and symptoms may occur along with or after CNS symptoms, and may include hypertension or hypotension, tachycardia or bradycardia, followed by ventricular arrhythmias and/or asystole.
- **Management** – LAST should be recognized and treated rapidly, prioritizing prevention of hypoxia and acidosis, seizure suppression, and cardiovascular support, as outlined in the Rapid Overview (table 4). (See 'Management of LAST' above.)
- **Lipid rescue** – We recommend administration of intravenous lipid emulsion (ILE) for patients with LAST who exhibit seizures or impending seizures, or signs of cardiovascular toxicity (ie, arrhythmias, severe hypotension, or cardiac arrest) along with ACLS (**Grade 1C**). Though the mechanism of action of lipid rescue and optional dose have yet to be determined, LAST may be fatal and adverse effects of lipid emulsion appear to be rare. Recommended doses of ILE are shown in the rapid overview (table 4). (See 'Lipid rescue' above.)

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Topic 90613 Version 32.0

GRAPHICS

Table 1: **ASRA preprocedure checklist**

1. Patient is identified, two criteria.
2. Allergies and anticoagulation status are reviewed.
3. Surgical procedure/consent is confirmed.
4. Block plan is confirmed, site is marked.
5. Necessary equipment is present, drugs/solutions are labeled.
6. Resuscitation equipment is immediately available: Airway devices, suction, vasoactive drugs, lipid emulsion.
7. Appropriate ASA* monitors are applied; intravenous access, sedation, and supplemental oxygen are provided, if indicated.
8. Aseptic technique is used: Hand cleansing is performed, mask and sterile gloves are used.
9. "Time out" is performed before needle insertion for each new block site if the position is changed or separated in time or performed by another team.

* Corrected based on erratum.[1]

Reference:

1. A checklist for performing regional nerve blocks: Erratum. Reg Anesth Pain Med 2014; 39:357.

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Table 2: **World Health Organization surgical safety checklist**

Sign in	Time out	Sign out
Before induction of anesthesia	**Before skin incision**	**Before patient leaves operating room**
☐ Patient has confirmed: ▪ Identity ▪ Site ▪ Procedure ▪ Consent ☐ Site marked/not applicable ☐ Anesthesia safety check completed ☐ Pulse oximeter on patient and functioning **Does patient have a:** Known allergy? ☐ No ☐ Yes Difficult airway/aspiration risk? ☐ No ☐ Yes, and equipment/assistance available Risk of >500 mL blood loss (7 mL/kg in children)? ☐ No ☐ Yes, and adequate intravenous access and fluids planned	☐ Confirm all team members have introduced themselves by name and role ☐ Surgeon, anesthesia professional, and nurse verbally confirm ▪ Patient ▪ Site ▪ Procedure **Anticipated critical events** ☐ Surgeon reviews: What are the critical or unexpected steps, operative duration, anticipated blood loss? ☐ Anesthesia team reviews: Are there any patient-specific concerns? ☐ Nursing team reviews: Has sterility (including indicator results) been confirmed? Are there equipment issues or any concerns? Has antibiotic prophylaxis been given within the last 60 minutes? ☐ Yes ☐ Not applicable Is essential imaging displayed? ☐ Yes ☐ Not applicable	**Nurse verbally confirms with the team:** ☐ The name of the procedure recorded ☐ That instrument, sponge, and needle counts are correct (or not applicable) ☐ How the specimen is labeled (including patient name) ☐ Whether there are any equipment problems to be addressed ☐ Surgeon, anesthesia professional, and nurse review the key concerns for recovery and management of this patient

This checklist is not intended to be comprehensive. Additions and modifications to fit local practice are encouraged.

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Table 3: **Local anesthetics commonly used for peripheral nerve block**

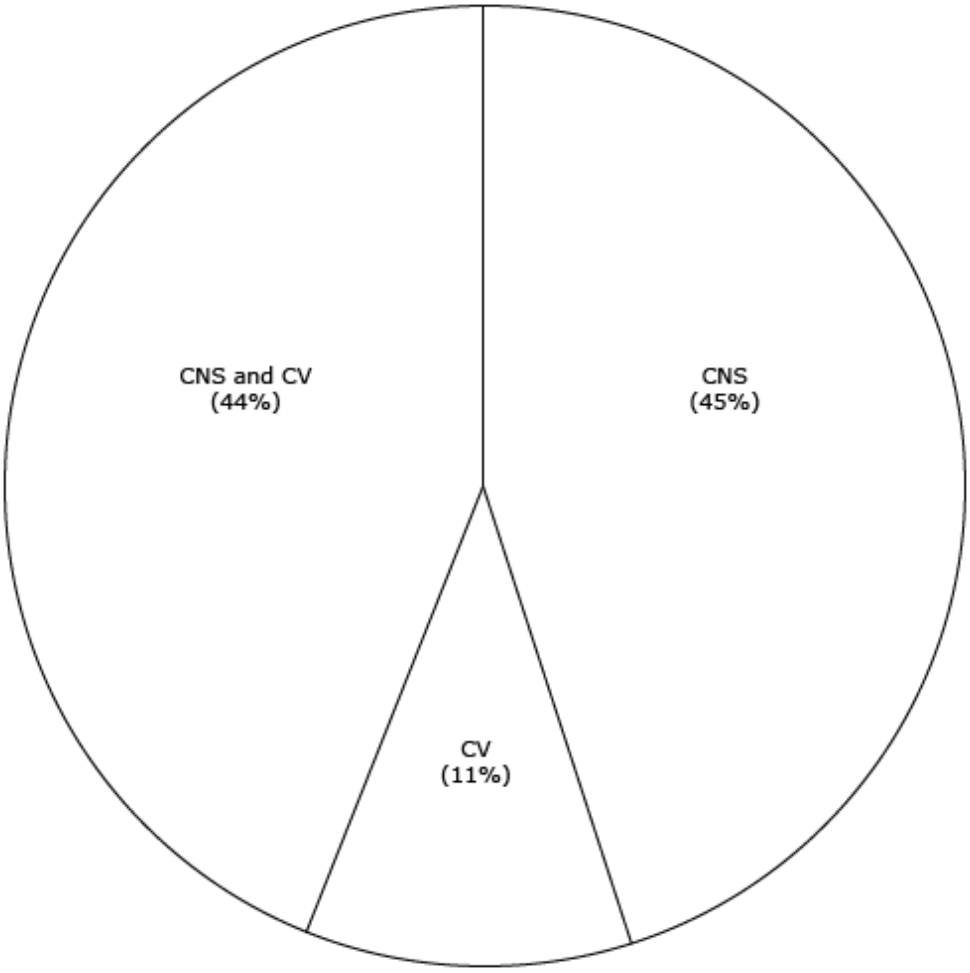
Anesthetic	Onset (minutes)	Duration of anesthesia* (hours)	Duration of analgesia* (hours)	Maximum dose*¶ (mg/kg) without/with epi
2% lidocaine	10 to 20	2 to 5	3 to 8	4.5/7
1.5% mepivacaine	10 to 20	2 to 5	3 to 10	5/7
0.2% ropivacaine	15 to 30	n/a	5 to 16	3/3.5
0.5% ropivacaine	15 to 30	4 to 12	5 to 24	3/3.5
0.25% bupivacaine	15 to 30	n/a	5 to 26	2.5/3
0.5% bupivacaine (+epi)	15 to 30	5 to 15	6 to 30	2.5/3

epi: epinephrine; n/a: not applicable.
* Duration varies widely by site of injection. These are generalized ranges of duration.
¶ Maximal doses represent general guidelines for tissue infiltration, nerve block, or epidural injection. Systemic toxicity may occur with doses below the recommended range, particularly with intravascular injection. Doses in excess of the recommended maximums have been administered without toxicity. These recommendations do not account for the site of injection, rate of administration, or the presence of risk factors for systemic toxicity (eg, renal or hepatic dysfunction, cardiac failure, pregnancy, or extremes of age).

Adapted from: Gadsen J. Local Anesthetics: Clinical Pharmacology and Rational Selection. The New York School of Regional Anesthesia website, October 2013.

Graphic 93203 Version 11.0

Figure 1: **Spectrum of presenting signs of LAST**



The frequency of symptoms and signs referable to CV, CNS, or both is given for the 93 cases in this review.

LAST: local anesthetic systemic toxicity; CV: cardiovascular; CNS: central nervous system.

Reproduced with permission from: Di Gregorio G, Neal JM, Rosenquist RW, Weinberg GL. Clinical presentation of local anesthetic systemic toxicity: A review of published cases, 1979 to 2009. Reg Anesth Pain Med 2010; 35(2)181-7. Copyright © 2010 BMJ Publishing Group Ltd.

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Figure 2: **Distribution of cardiovascular manifestations of LAST**

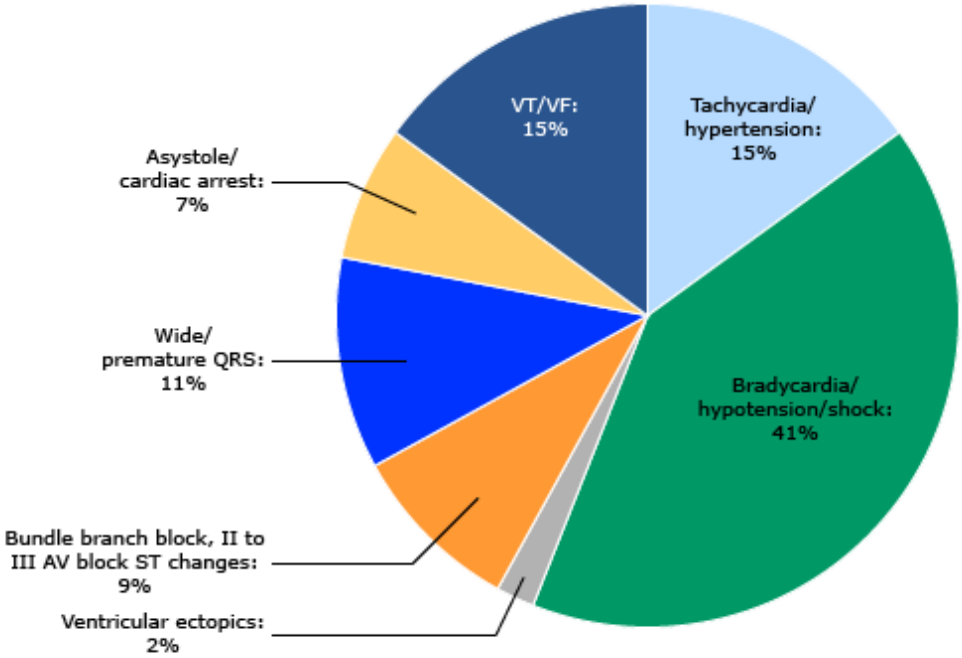

Diagram showing the prevalence of different cardiovascular manifestations of toxicity in LAST cases reported between 2010 and 2014.

LAST: local anesthetic systemic toxicity; VT/VF: ventricular tachycardia/ventricular fibrillation; AV: atrioventricular; ST: ST tract.

Reproduced with permission from: Vasques F, Behr AU, Weinberg G, et al. A review of local anesthetic systemic toxicity cases since publication of the American Society of Regional Anesthesia recommendations: To whom it may concern. Reg Anesth Pain Med 2015; 40(6):698-705. Copyright © 2015 BMJ Publishing Group Ltd.

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Figure 3: **Distribution of central nervous system manifestations of LAST**

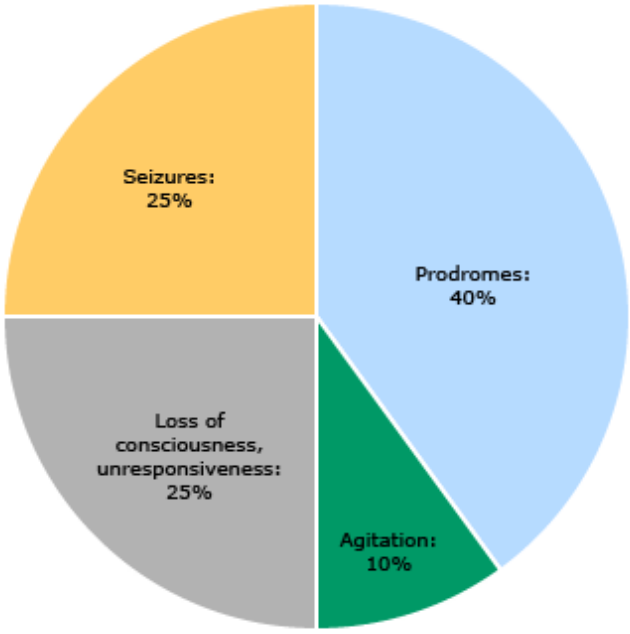

Diagram showing the prevalence of different neurological manifestations of toxicity in the LAST cases reported between 2010 and 2014.

LAST: local anesthetic systemic toxicity.

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Figure 4: **Local anesthetics in reported cases of LAST**

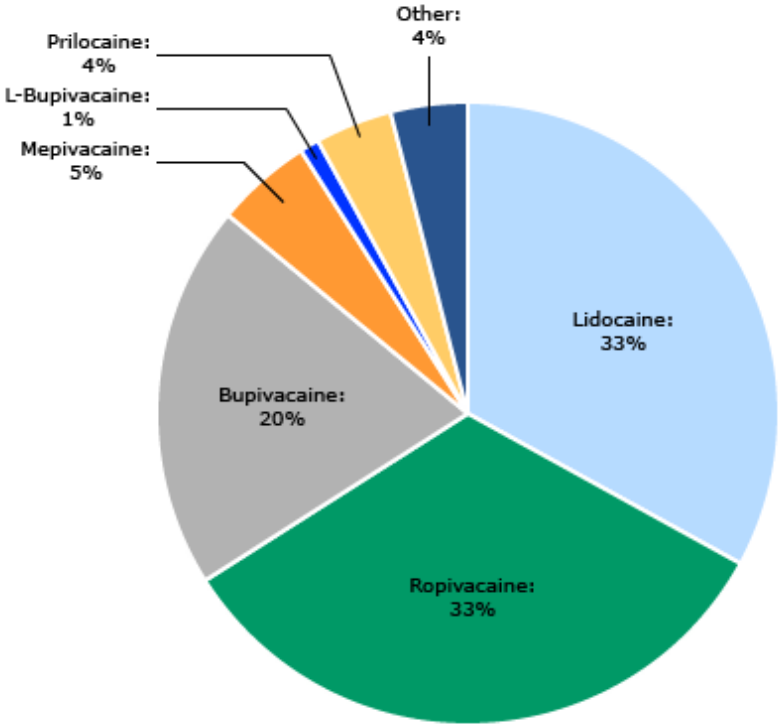

Diagram showing the proportion of local anesthetic drugs used in LAST cases reported between 2010 and 2014.

LAST: local anesthetic systemic toxicity.

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Figure 5: **Onset time in reported cases of LAST**

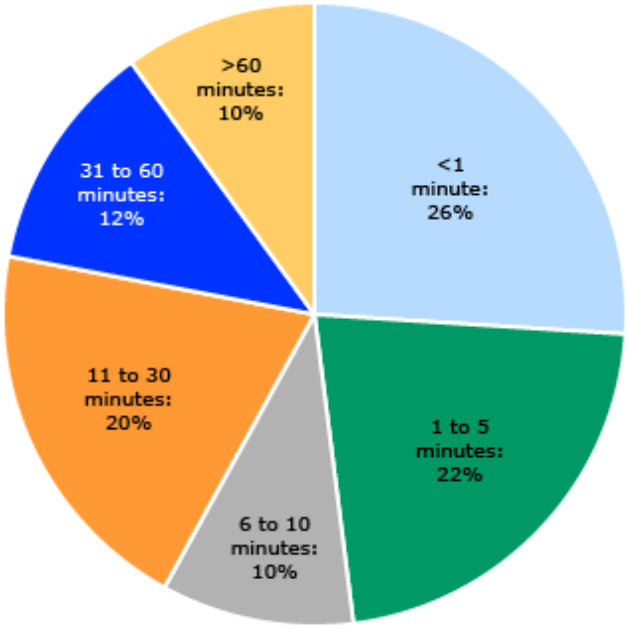

Diagram showing the onset after single-shot regional anesthesia in LAST cases reported between 2010 and 2014.

LAST: local anesthetic systemic toxicity.

Reproduced with permission from: Vasques F, Behr AU, Weinberg G, et al. A review of local anesthetic systemic toxicity cases since publication of the American Society of Regional Anesthesia recommendations: To whom it may concern. Reg Anesth Pain Med 2015; 40(6):698-705. Copyright © 2015 BMJ Publishing Group Ltd.

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Table 4: **Local anesthetic systemic toxicity: Rapid overview** [1]

Etiology
▪ Unintended intravascular injection of LA ▪ Systemic absorption of LA
Risk factors
▪ Extremes of age (ie, <4 months or >79 years) ▪ Cardiac conduction disease ▪ Ischemic heart disease ▪ Renal dysfunction ▪ Hepatic dysfunction ▪ Pregnancy ▪ Carnitine deficiency ▪ Highly vascular block site
Signs and symptoms* LAST can occur >15 minutes after injection of LA
▪ CNS:¶ • Tinnitus • Circumoral numbness • Metallic taste • Agitation • Dysarthria • Seizures • Loss of consciousness • Respiratory arrest ▪ Cardiovascular:¶ • Hypotension • Bradycardia • Ventricular arrhythmias • Cardiovascular collapse
Treatment*
▪ Stop injection or infusion ▪ Call for help and lipid emulsion ▪ While stabilizing the patient, arrange for cardiopulmonary bypass ▪ Airway management: Ventilate with 100% oxygen, prevent hypoxemia, hypercarbia, and acidosis ▪ Suppress seizures: Benzodiazepines preferred (avoid large doses of propofol) ▪ Manage arrhythmias and cardiac arrest per ACLS **EXCEPT**: • **Reduce** individual epinephrine boluses to ≤1 mcg/kg • **Avoid** vasopressin, calcium channel blockers, beta blockers, and LA • Administer amiodarone as the first line antiarrhythmic ▪ Institute lipid emulsion therapy with 20% lipid emulsion

- Adults ≥70 kg: Bolus 100 mL IV over 2 to 3 minutes, followed by infusion of 250 mL over 15 to 20 minutes
- Children or adults <70 kg: Bolus 1.5 mL/kg IBW IV over 2 to 3 minutes, followed by infusion at 0.25 mL/kg/minute
- For patients with persistent cardiovascular instability, repeat bolus and double infusion rate
- Continue infusion for at least 15 minutes after hemodynamic stability is achieved
- Maximum dose lipid emulsion approximately 12 mL/kg IV
- **Note**: Propofol is not a substitute for lipid emulsion

- Institute cardiopulmonary bypass for LAST unresponsive to lipid emulsion and ACLS
- Monitor the patient for 2 hours after seizure, 4 to 6 hours after hemodynamic instability, or as appropriate after cardiac arrest

For further information, refer to UpToDate content on local anesthetic systemic toxicity.

ACLS: advanced cardiac life support; CNS: central nervous system; IBW: ideal body weight; IV: intravenous; LA: local anesthetic; LAST: local anesthetic systemic toxicity; TPN: total parenteral nutrition.

* Not all signs and symptoms occur in every patient.

¶ Cardiovascular collapse may occur without CNS signs or symptoms, especially with rapid intravascular injection.

Reference:

1. *Neal JM, Barrington MJ, Fettiplace MR, et al. The Third American Society of Regional Anesthesia and Pain Medicine Practice Advisory on Local Anesthetic Systemic Toxicity: Executive Summary 2017. Reg Anesth Pain Med 2018; 43:113.*
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Graphic 107887 Version 9.0

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